

Ashwin Badamikar

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EDUCATION

Northeastern University

MS Information Systems | GPA: 3.5/4.0

Sep 2024 - Dec 2026

Boston, MA

JSS Academy of Technical Education

BE Computer Science and Engineering

Aug 2019 - May 2023

Bengaluru, India

EXPERIENCE

Aosenuma Inc.

AI Engineer Intern

Jan 2026 - Jun 2026

Remote, USA

- Cut ESG document review time significantly by building **StakeAI**, a production multi agent research automation system in LangGraph that orchestrated entity extraction, knowledge graph generation, and structured report writing, giving analysts natural language access to over 500 daily document outputs with zero regression delivery across three live production services running around the clock without manual intervention
- Reduced hallucination rates by **22%** and cut model drift detection time by **95%** on **RapidAI**, an agentic evaluation pipeline built across 4 LLM configurations, by designing automated eval workflows and applying prompt engineering fixes informed directly by the results those evals produced
- Deployed **WhatsApp Summary Bot** as a multi tenant enterprise chatbot by integrating Pinecone vector memory with namespace isolation for secure per tenant context separation, then instrumenting Helicone observability to track token cost, latency, and error rates across every production interaction

TurnOn

Software Engineer

Jan 2023 - Jul 2024

Bengaluru, India

- Delivered **Bhanush**, an agricultural IoT analytics platform, by building Kafka streaming pipelines partitioned by device ID and time window to preserve temporal coherence for downstream models, improving throughput by **20%** across more than 100,000 daily sensor events without sacrificing schema consistency or long term data integrity across the full pipeline architecture and deployment and ongoing maintenance
- Accelerated client decisions by **15%** on **Smart Build Automation**, a construction analytics platform, by shipping ETL pipelines with **Pydantic schema validation** and real time Power BI dashboards that surfaced risk patterns and anomalies before stakeholders had to ask
- Built a single **auditable source of truth** for two client verticals by replacing ad hoc SQL transformations with governed Airflow, Snowflake, and dbt pipelines, improving business forecasting accuracy by 15 percent while enabling full compliance audit trails

JSS Academy of Technical Education

Research Assistant

Sep 2022 - May 2023

Bengaluru, India

- Earned national recognition from the **Karnataka State Council for Science and Technology** for leading **Cross Lingual Transfer Learning for NLI** across 15 language pairs on the XNLI benchmark, choosing multilingual BERT fine tuning because transfer learning outperformed task specific models
- Improved reproducibility across a 4 person research team on the same NLI project by reviewing code and evaluation pipelines every week, then translating the findings into **peer reviewed reports** that were clear to both faculty and non specialist external reviewers

PROJECTS

Chronic Care AI - Agentic Healthcare System

Python · LangGraph · FastAPI · Pinecone · Snowflake · Airflow

- Shipped a **HIPAA compliant multi agent system** using LangGraph agents for symptom analysis, medication lookup, and care coordination, achieving a **28% improvement in retrieval precision** with Mistral embeddings over Pinecone semantic memory search
- Deployed the full stack via **Docker and GitHub Actions CI/CD** on a FastAPI and Snowflake backend, integrating Google Maps and Edamam APIs as clean agent accessible tools for geolocation aware care routing and nutrition informed treatment planning decisions

Financial Statement Data Pipeline - SEC Automation

Python · Snowflake · Airflow · dbt · FastAPI

- Replaced a fully manual analyst workflow by automating ingestion of **500 plus SEC financial disclosures** through a multi stage Snowflake, Airflow, and dbt pipeline enforced end to end with **Pydantic data contracts** and rigorous 3NF data modeling standards across every stage of the ingestion process
- Built a **self service FastAPI and Streamlit interface** that let analysts extract financial insights independently, cutting their dependency on engineering support for routine data requests and one off reporting asks across teams and departments

Nutrition Tutor Bot - Multimodal RAG Agent

Python · GPT-4 Vision · ChromaDB · LangChain · Streamlit

- Reduced hallucination rate by **22%** across 1,000 plus queries by combining GPT 4 Vision multimodal food image analysis with a hybrid dense and sparse retrieval architecture grounded directly in eval results rather than framework defaults or generic prompt tuning alone
- Integrated the USDA Nutrition API as a **live grounded data source** and deployed the full Streamlit application, validating response accuracy with real user queries before every production release across all supported platforms and device types without exception

NBA RAG System - Conversational Sports Intelligence

PostgreSQL · pgvector · Ollama · FastAPI · Angular

- Built an end to end **RAG pipeline with pgvector** for dense embedding storage and offline local LLM inference through Ollama, then wrapped it in Angular chat interface with real time streaming for **sub second natural language Q&A** over player statistics and team performance data

SKILLS

AI, Agents and Models: LangChain, LangGraph, CrewAI, RAG, Agentic RAG, MCP, Multi Agent Orchestration, Evals, Guardrails, Prompt Engineering, Fine Tuning, GPT 4, GPT 4 Vision, Claude Sonnet, Mistral AI, Azure OpenAI, Amazon Bedrock, Hugging Face Transformers, PyTorch

Languages, Frameworks and Data Engineering: Python, SQL, TypeScript, JavaScript, FastAPI, Flask, Node.js, React, Next.js, REST APIs, GraphQL, Pydantic, Pytest, Pinecone, ChromaDB, pgvector, FAISS, Airflow, dbt, Kafka, Spark, Snowflake, PostgreSQL, Databricks, BigQuery, Redis, MongoDB

Cloud and DevOps: AWS (S3, Lambda, Bedrock), Azure, GCP, Docker, Kubernetes, GitHub Actions, Terraform, Jenkins, CI/CD, Helicone, MLflow, Git

Certifications: Generative AI (LinkedIn) | Quantum-Safe Cryptography (IBM) | Cloud Computing Architecture (Great Learning)